DT

D A V I D K. T A N

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CLEARANCE

Fully Adjudicated TS/SAP | Active Secret Clearance

TECHNICAL SKILLS

- Programming: C++, Python, Java.
- Software Engineering: Object-Oriented Design (OOD), Agile/Scrum.
- DevSecOps: GitLab CI/CD, Jenkins, Kubernetes, Docker.
- System & Cloud: Linux, AWS, Red Hat Enterprise Linux (RHEL).
- Testing & Validation: Unit Testing, Integration Testing, and Debugging.
- Modeling & Simulation: 6-DOF Flight Dynamics, Missile Simulation and Monte Carlo Analysis.

PROFESSIONAL EXPERIENCE

Software Design Engineer III, **Torch Technologies**, Feb 2026 - Current

- Developed C++ and Python features within model development and simulation execution frameworks, integrating updates across three software components and streamlining debugging workflows used for model validation.
- Diagnosed and resolved complex C++ and Python software defects using Linux, Visual Studio, Git, and automated unit and integration testing, reducing recurring issues by approximately 10% and improving software stability.
- Translated system requirements into software designs, code implementations, and verification documentation, reducing integration rework by approximately 15% and improving delivery efficiency.
- Collaborated with software, systems, and simulation engineers to design, integrate, and validate software enhancements across shared simulation baselines, ensuring compatibility and reliable end-to-end functionality.
- Designed, implemented, and validated high-fidelity C++ six-degree-of-freedom (6-DOF) missile simulation models, including autopilot control logic and the integration of guidance, sensor, and flight-dynamics components for LRPM and Hydra-70, increasing simulation throughput by 10%.

Software Engineer II, **BAE Systems** , Sep 2024 - Jan 2026, Utah

- Developed, reviewed, and tested C++11 six-degree-of-freedom (6-DOF) flight-dynamics models supporting LGM-35A Sentinel ICBM boost and post-boost performance, maintaining 100% traceability between software implementation and flight-performance requirements.
- Diagnosed and resolved simulation inconsistencies by analyzing flight behavior, debugging C++ model logic, and refining model assumptions, improving model fidelity and reducing downstream verification and validation effort.
- Implemented and validated software updates through code reviews, regression testing, and requirements-based analysis, helping maintain reliable simulation performance across evolving model baselines.

Software Engineer I, **Space Dynamics Laboratory**, Jan 2023 - Sep 2024, Utah

- Developed and maintained C++ and Python software modules within the FORGE MDPAF ground framework, implementing enhancements, debugging issues, and supporting reliable data processing capabilities.
- Collaborated with cross-functional engineering teams to analyze requirements, troubleshoot software defects, and deliver high-quality solutions that improved system reliability and performance.

Information Technology Specialist, **Air Force**, Jan 2019 - Dec 2022

- Provided technical support for 100+ users, troubleshooting hardware, software, and network issues.
- Installed, configured, and maintained 100+ workstations help ensuring 98% system availability and reliable support for daily military operations.

PROJECTS

- Mission Planning Dashboard: Built a full-stack mission-planning dashboard with role-based workflows, REST API endpoints, and audit logging, reducing planning data-entry errors by 30%.
- Missile Flight Dynamics Simulation: Built a standalone C++ 6-DOF missile simulation with autopilot control logic, guidance interfaces, and flight-dynamics behavior; validated model performance through Monte Carlo analysis.

EDUCATION

Master of Science, Aerospace Engineering — Dec 2033
Colorado State University

Master of Science, Computer Science — May 2027
Georgia Institute of Technology

Bachelor of Science, Computer Science — Dec 2022
Valdosta State University

CERTIFICATIONS

CISSP | CKA | CKS | AWS Solutions Architect | Security+ | PMP

PUBLICATION

- Co-authored peer-reviewed Springer publication on distributed fault-containment simulation using randomized scheduling, published in *Advanced Communication and Intelligent Systems.*